

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
```

```
df=pd.read_csv("/content/mutual_funds_data.csv")
```

```
df.head()
```

	scheme_name	min_sip	min_lumpsum	expense_ratio	fund_size_cr	fund_age_yr	fund_manager	sortino	alpha	sd	beta	sh
0	Aditya Birla SL Active Debt Multi- Mgr FoF-Dir ...	100	100	0.27	10.0	10	Kaustubh Gupta	0.32	2.24	9.39	0.01	
1	Aditya Birla SL Arbitrage Fund	1000	1000	0.36	4288.0	10	Lovelish Solanki	1.33	1.53	0.72	0.56	
2	Aditya Birla SL Asset Allocator	1000	1000	0.53	157.0	10	Vinod Narayan Bhat	3.44	2.67	10.58	0.67	

```
df.shape
```

Show hidden output

```
df.columns
```

```
Index(['scheme_name', 'min_sip', 'min_lumpsum', 'expense_ratio',  
      'fund_size_cr', 'fund_age_yr', 'fund_manager', 'sortino', 'alpha', 'sd',  
      'beta', 'sharpe', 'risk_level', 'amc_name', 'rating', 'category',  
      'sub_category', 'returns_1yr', 'returns_3yr', 'returns_5yr'],  
      dtype='object')
```

```
df.describe()
```

	min_sip	min_lumpsum	expense_ratio	fund_size_cr	fund_age_yr	risk_level	rating	returns_1yr	returns_3yr
count	814.000000	814.000000	814.000000	814.000000	814.000000	814.000000	814.000000	814.000000	793.000000
mean	528.500000	3047.468059	0.713231	3812.854214	8.319410	4.400491	2.641278	3.921376	18.525347
std	369.184517	2514.619513	0.482161	7181.479475	2.640622	1.814741	1.463476	6.675502	12.108476
min	0.000000	0.000000	0.000000	2.380000	1.000000	1.000000	0.000000	-19.700000	3.300000
25%	150.000000	500.000000	0.310000	175.250000	6.000000	3.000000	2.000000	1.500000	6.200000
50%	500.000000	5000.000000	0.615000	940.500000	10.000000	6.000000	3.000000	4.400000	18.500000
75%	1000.000000	5000.000000	1.030000	3819.000000	10.000000	6.000000	4.000000	5.600000	27.100000
max	2000.000000	25000.000000	2.590000	57052.000000	17.000000	6.000000	5.000000	130.800000	71.400000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 814 entries, 0 to 813
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   scheme_name           814 non-null   object
1   min_sip               814 non-null   int64
2   min_lumpsum           814 non-null   int64
3   expense_ratio         814 non-null   float64
4   fund_size_cr          814 non-null   float64
5   fund_age_yr           814 non-null   int64
6   fund_manager          814 non-null   object
7   sortino               814 non-null   object
8   alpha                814 non-null   object
9   sd                   814 non-null   object
10  beta                 814 non-null   object
11  sharpe               814 non-null   object
12  risk_level            814 non-null   int64
```

```
13  amc_name      814 non-null  object
14  rating        814 non-null  int64
15  category      814 non-null  object
16  sub_category  814 non-null  object
17  returns_1yr   814 non-null  float64
18  returns_3yr   793 non-null  float64
19  returns_5yr   647 non-null  float64
dtypes: float64(5), int64(5), object(10)
memory usage: 127.3+ KB
```

df.dtypes

	0
scheme_name	object
min_sip	int64
min_lumpsum	int64
expense_ratio	float64
fund_size_cr	float64
fund_age_yr	int64
fund_manager	object
sortino	object
alpha	object
sd	object
beta	object
sharpe	object
risk_level	int64
amc_name	object
rating	int64
category	object
sub_category	object
returns_1yr	float64
returns_3yr	float64
returns_5yr	float64

dtype: object

df.nunique()

```
0
scheme_name 789
min_sip      10
min_lumpsum  11
expense_ratio 177
fund_size_cr 693
fund_age_yr  14
fund_manager 260
sortino      385
alpha        540
sd           581
beta         162
sharpe       217
risk_level   6
amc_name     39
rating       6
category     5
sub_category 38
returns_1yr  210
returns_3yr  304
returns_5yr  147
```

dtype: int64

```
df.isnull().sum()
```

```
0
scheme_name 0
min_sip      0
min_lumpsum  0
expense_ratio 0
fund_size_cr 0
fund_age_yr  0
fund_manager 0
sortino      0
alpha        0
sd           0
beta         0
sharpe       0
risk_level   0
amc_name     0
rating       0
category     0
sub_category 0
returns_1yr  0
returns_3yr  21
returns_5yr  167
```

dtype: int64

```
pe=(df.isnull().sum()/len(df))*100
```

Start coding or generate with AI.

```
df["returns_3yr"] = df["returns_3yr"].fillna(df["returns_3yr"].mean())
```

```
df['returns_5yr']=df['returns_5yr'].fillna(df['returns_5yr'].mean())
```

```
df.isnull().sum()
```

	0
scheme_name	0
min_sip	0
min_lumpsum	0
expense_ratio	0
fund_size_cr	0
fund_age_yr	0
fund_manager	0
sortino	0
alpha	0
sd	0
beta	0
sharpe	0
risk_level	0
amc_name	0
rating	0
category	0
sub_category	0
returns_1yr	0
returns_3yr	0
returns_5yr	0

dtype: int64

```
df.duplicated().sum()
```

np.int64(0)

```
df.head()
```

	scheme_name	min_sip	min_lumpsum	expense_ratio	fund_size_cr	fund_age_yr	fund_manager	sortino	alpha	sd	beta	sh
0	Aditya Birla SL Active Debt Multi- Mgr FoF-Dir ...	100	100	0.27	10.0	10	Kaustubh Gupta	0.32	2.24	9.39	0.01	
1	Aditya Birla SL Arbitrage Fund	1000	1000	0.36	4288.0	10	Lovelish Solanki	1.33	1.53	0.72	0.56	
2	Aditya Birla SL Asset Allocator	1000	1000	0.53	157.0	10	Vinod Narayan Bhat	3.44	2.67	10.58	0.67	

```
numeric_cols=df.select_dtypes(include=np.number).columns.tolist()
print(numeric_cols)
```

```
['min_sip', 'min_lumpsum', 'expense_ratio', 'fund_size_cr', 'fund_age_yr', 'risk_level', 'rating', 'returns_1yr', 'returns_3yr', 'returns_5yr']
```

```
for col in numeric_cols:
    df[col]=(df[col].
        astype(str)
        .str.replace('%','',regex=False)
        .str.replace(",","",regex=False)
        )
    df[col]=pd.to_numeric(df[col],errors="coerce")
    df[col]=df[col].fillna(df[col].median())
```

```
df.dtypes
```

	0
scheme_name	object
min_sip	int64
min_lumpsum	int64
expense_ratio	float64
fund_size_cr	float64
fund_age_yr	int64
fund_manager	object
sortino	object
alpha	object
sd	object
beta	object
sharpe	object
risk_level	int64
amc_name	object
rating	int64
category	object
sub_category	object
returns_1yr	float64
returns_3yr	float64
returns_5yr	float64

```
dtype: object
```

```
df.isnull().sum()
```

```
0
scheme_name 0
min_sip     0
min_lumpsum 0
expense_ratio 0
fund_size_cr 0
fund_age_yr 0
fund_manager 0
sortino     0
alpha       0
sd          0
beta        0
sharpe      0
risk_level  0
amc_name    0
rating      0
category    0
sub_category 0
returns_1yr 0
returns_3yr 0
returns_5yr 0
```

dtype: int64

```
df.fillna(df.median(numeric_only=True), inplace=True)
```

```
df.fillna("Unknown", inplace=True)
```

```
df["category"].value_counts()
```

	count
Equity	308
Debt	282
Hybrid	116
Other	80
Solution Oriented	28

dtype: int64

```
df['amc_name'].value_counts()
```

	count
amc_name	
ICICI Prudential Mutual Fund	57
Aditya Birla Sun Life Mutual Fund	49
SBI Mutual Fund	48
Sundaram Mutual Fund	42
Nippon India Mutual Fund	39
HDFC Mutual Fund	38
UTI Mutual Fund	37
HSBC Mutual Fund	35
Axis Mutual Fund	35
DSP Mutual Fund	35
Tata Mutual Fund	34
Kotak Mahindra Mutual Fund	32
Invesco Mutual Fund	27
Franklin Templeton Mutual Fund	26
Baroda BNP Paribas Mutual Fund	24
LIC Mutual Fund	20
Edelweiss Mutual Fund	20
Canara Robeco Mutual Fund	18
IDBI Mutual Fund	18
PGIM India Mutual Fund	18
Mirae Asset Mutual Fund	18
Union Mutual Fund	16
Motilal Oswal Mutual Fund	16
Bank of India Mutual Fund	15
Mahindra Manulife Mutual Fund	14
JM Financial Mutual Fund	11
Quant Mutual Fund	11
Indiabulls Mutual Fund	9
ITI Mutual Fund	9
Navi Mutual Fund	8

```
df['expense_ratio'].min()
```

0.0	Quantum Mutual Fund	7
	Shriram Mutual Fund	4

```
df["returns_5yr"].max()
```

23.2	IIFL Mutual Fund	3
	WhiteOak Capital Mutual Fund	3

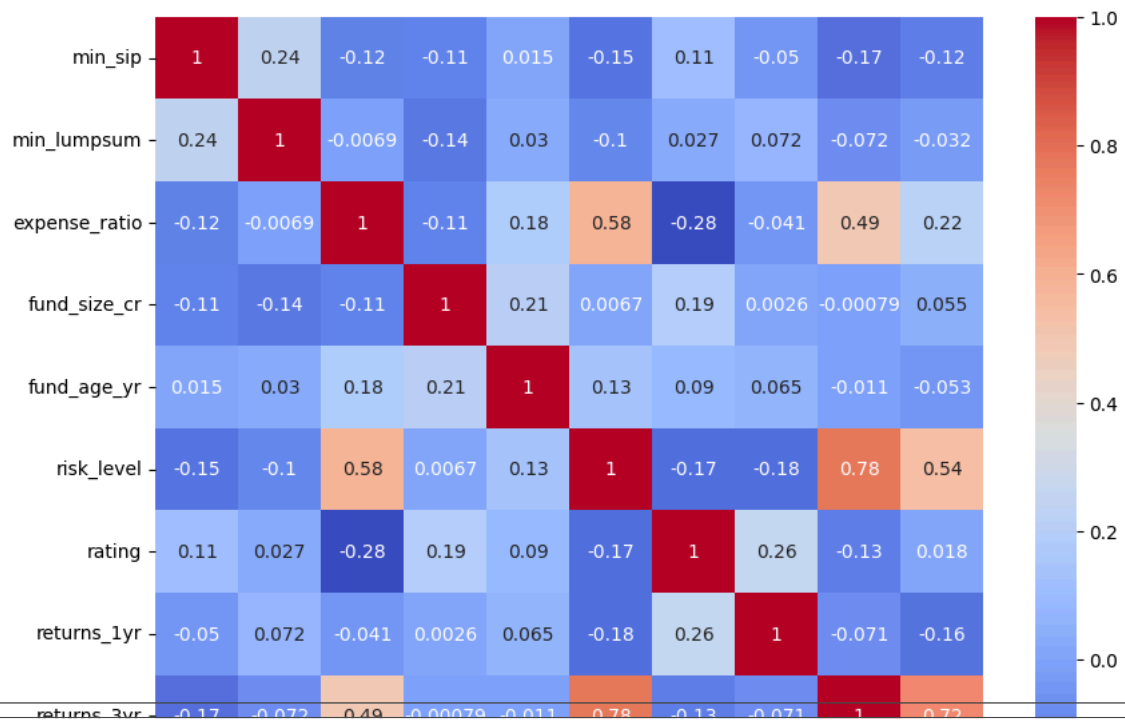
```
df.corr(numeric_only=True)
```

	Bandhan Mutual Fund	2
	L&T Mutual Fund	2

```
dtype: int64
```

	min_sip	min_lumpsum	expense_ratio	fund_size_cr	fund_age_yr	risk_level	rating	returns_1yr	returns_3yr	returns_5yr
min_sip	1.000000	0.236304	-0.119078	-0.111495	0.014810	-0.151352	0.111263	-0.049756	-0.167391	-0.123567
min_lumpsum	0.236304	1.000000	-0.006859	-0.142937	0.030491	-0.102806	0.026659	0.072438	-0.072172	-0.031815
expense_ratio	-0.119078	-0.006859	1.000000	-0.109883	0.180308	0.580970	-0.279715	-0.041015	0.489621	0.215298
fund_size_cr	-0.111495	-0.142937	-0.109883	1.000000	0.211195	0.006744	0.194323	0.002586	-0.000791	0.054535
fund_age_yr	0.014810	0.030491	0.180308	0.211195	1.000000	0.133440	0.090160	0.065085	-0.011475	-0.053500
risk_level	-0.151352	-0.102806	0.580970	0.006744	0.133440	1.000000	-0.170460	-0.176473	0.778559	0.542423
rating	0.111263	0.026659	-0.279715	0.194323	0.090160	-0.170460	1.000000	0.260413	-0.129449	0.017925
returns_1yr	-0.049756	0.072438	-0.041015	0.002586	0.065085	-0.176473	0.260413	1.000000	-0.071361	-0.160945
returns_3yr	-0.167391	-0.072172	0.489621	-0.000791	-0.011475	0.778559	-0.129449	-0.071361	1.000000	0.722586
returns_5yr	-0.123567	-0.031815	0.215298	0.054535	-0.053500	0.542423	0.017925	-0.160945	0.722586	1.000000

```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap="coolwarm")
plt.show()
```



```
df.columns
```

```
Index(['fund_size_cr', 'fund_age_yr', 'fund_manager', 'sortino', 'alpha', 'sd',
      'beta', 'sharpe', 'risk_level', 'amc_name', 'rating', 'category',
      'sub_category', 'returns_1yr', 'returns_3yr', 'returns_5yr'],
      dtype='object', name='columns')
```

```
features = [
    'expense_ratio',
    'returns_1yr',
    'returns_3yr',
    'returns_5yr',
    'sharpe',
    'sortino',
    'alpha',
    'beta'
]

score_df = df[features].copy()
score_df.head()
```

	expense_ratio	returns_1yr	returns_3yr	returns_5yr	sharpe	sortino	alpha	beta
0	0.27	4.0	6.5	6.900000	0.24	0.32	2.24	0.01
1	0.36	5.6	4.8	5.500000	1.1	1.33	1.53	0.56
2	0.53	2.0	18.9	9.700000	1.42	3.44	2.67	0.67
3	0.76	-0.7	17.1	9.490726	0.9	2.18	-6.37	0.85
4	0.61	4.5	18.6	9.700000	1.39	3.69	1.99	0.68

```
score_df.isnull().sum()
```

```

      0
expense_ratio  0
returns_1yr    0
returns_3yr    0
returns_5yr    0
sharpe        0
sortino       0
alpha        0
beta         0

```

```
dtype: int64
```

```
score_df = score_df.apply(pd.to_numeric, errors='coerce')
```

```
score_df.dtypes
```

```

      0
expense_ratio  float64
returns_1yr    float64
returns_3yr    float64
returns_5yr    float64
sharpe        float64
sortino       float64
alpha        float64
beta         float64

```

```
dtype: object
```

```

scaler=MinMaxScaler()
normalized=scaler.fit_transform(score_df)

```

```

normalized = pd.DataFrame(
normalized,
columns=features
)

```

```
score_df
```

	expense_ratio	returns_1yr	returns_3yr	returns_5yr	sharpe	sortino	alpha	beta
0	0.27	4.0	6.5	6.900000	0.24	0.32	2.24	0.01
1	0.36	5.6	4.8	5.500000	1.10	1.33	1.53	0.56
2	0.53	2.0	18.9	9.700000	1.42	3.44	2.67	0.67
3	0.76	-0.7	17.1	9.490726	0.90	2.18	-6.37	0.85
4	0.61	4.5	18.6	9.700000	1.39	3.69	1.99	0.68
...	...	...	...	...	...	...	...	...
809	0.96	-0.7	13.1	6.200000	1.29	2.47	0.92	0.44
810	1.16	1.7	27.2	12.200000	1.48	3.21	2.73	0.90
811	0.26	5.3	3.8	9.490726	-0.07	-0.42	0.93	0.72
812	0.22	1.7	32.4	13.300000	-0.07	-0.42	0.93	0.72
813	0.33	5.3	4.5	9.490726	0.80	2.81	1.20	0.47

814 rows × 8 columns

```
normalized['expense_ratio'] = 1-normalized['expense_ratio']
```

```
normalized['beta']=1-normalized['beta']
```

```
normalized['Composite Score'] = (
    normalized['returns_1yr'] * 0.20 +
    normalized['returns_3yr'] * 0.20 +
    normalized['returns_5yr'] * 0.20 +
    normalized['expense_ratio'] * 0.15 +
    normalized['sharpe'] * 0.10 +
    normalized['sortino'] * 0.05 +
    normalized['alpha'] * 0.05 +
    normalized['beta'] * 0.05
)
```

```
df['Composite Score'] = normalized['Composite Score']
```

```
df['Rank']=df['Composite Score'].rank(
    ascending=False,
    method='dense'
)
```

```
df=df.sort_values(
    'Composite Score',
    ascending=False
)
```

```
top10=df.head(10)
```

```
top20=df.head(20)
```

```
top30=df.head(30)
```

```
df.replace('-', np.nan, inplace=True)
```

```
df.fillna(df.median(numeric_only=True), inplace=True)
```

Start coding or generate with AI.

```
df.isnull().sum()
```

	0
scheme_name	0
min_sip	0
min_lumpsum	0
expense_ratio	0
fund_size_cr	0
fund_age_yr	0
fund_manager	0
sortino	23
alpha	42
sd	24
beta	42
sharpe	23
risk_level	0
amc_name	0
rating	0
category	0
sub_category	0
returns_1yr	0
returns_3yr	0
returns_5yr	0
Composite Score	0

Rank 0

```
for col in ['sortino', 'alpha', 'sd', 'beta', 'sharpe']:  
    df[col] = pd.to_numeric(df[col], errors='coerce')  
df.fillna(df.median(numeric_only=True), inplace=True)
```

```
df.isnull().sum()
```

	0
scheme_name	0
min_sip	0
min_lumpsum	0
expense_ratio	0
fund_size_cr	0
fund_age_yr	0
fund_manager	0
sortino	0
alpha	0